ProofGate

Brand & Compliance Clearance System with Selective Revalidation

Hackathon: CodeNexus India 2026

Team: MANTIS

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Live Product: https://proof-gate-dusky.vercel.app/

Repository: https://github.com/Tanish9022/ProofGate

1. Executive Summary & Core Division of Labour

Marketing teams produce collateral at high velocity, but marketing compliance is notoriously prone to hallucinations, false positives, and stale clearances. Pure AI checkers hallucinate justifications, while naive pattern matching flags historical quotes as violations.

ProofGate resolves this with a strict architectural division of labour:

- **AI Proposes:** Gemini parses rich documents (TXT, PDF, DOCX, images), extracts factual claims with exact verbatim string quotes, labels surrounding context (`current`, `historical`, `quoted`, `hypothetical`), and localizes visual regions on uploaded assets.
- **Code Decides:** Deterministic TypeScript code strictly calculates numerical equality, currency rates, date ranges, annualised pricing, banned phrase regexes, and WCAG 2.1 relative luminance/contrast ratios. No generative model ever decides whether a numeric fact or colour contrast passes.
- **Humans Approve:** Reviewers make all clearance and override decisions. Every approval creates dependency rows pointing at exact source versions.
- **Selective Revalidation:** When approved facts change in the library, ProofGate traces dependencies and reopens only clearances with confirmed links to the updated source. Historical mentions remain safely cleared.

2. Complete Working Feature List

- Isolated per-visitor demo workspaces with zero-login evaluation via session cookies
- Built-in 5-minute interactive guided walkthrough tour banner across all core workflows
- Role switcher (Creator, Reviewer, Admin) with server-side authorization enforcement (403 on privileged actions)
- Multi-format content ingestion supporting raw text, PDF, DOCX, PNG, and JPEG documents
- Hybrid claim extraction with verbatim substring verification and context labeling (current, historical, quoted, hypothetical)
- Multi-model Gemini fallback cascade (gemini-2.0-flash → gemini-1.5-flash → gemini-1.5-flash-8b) with deterministic regex fallback
- Deterministic rules checker for numerical tolerances, unit conversions, date validation, and banned terminology
- In-document highlight inspector with side-by-side interactive finding cards and 1-click "Apply fix" replacements
- Creator override flow ("Dismiss with reason") requiring documented justification logged to audit trail
- Collaborative commenting on content versions with role attribution and audit logging
- Independent human sign-off review queue with transactional approval and rejection
- Idempotent clearance creation with dependency rows pointing at exact source versions
- Printable clearance certificate view with locked dependency manifest
- Source of truth library with slide-over drawer, multi-version history, and pre-publish impact analysis
- Source version lifecycle operations: supersede, withdraw, and expire
- Selective revalidation engine that re-opens only clearances with confirmed dependencies on modified sources
- Side-by-side Old Value vs. New Value diff view for revalidating updated copy
- "Not Affected" transparency list proving historical mentions remain safely cleared
- Context-aware dependency linking isolating ambiguous or qualified claims into a "Needs Confirming" probable-links queue
- Gold benchmark evaluation suite (50 cases) with real-time /evaluation dashboard reporting measured precision, recall, and F1

- Image brand and accessibility checking pipeline with Gemini vision region localization and deterministic WCAG AA contrast/palette distance verification
- Interactive SVG image overlay with colour-coded bounding boxes and click-to-inspect finding details
- Governance insights dashboard (/insights) with 6 core operational metrics and automated unclear-rule detection (≥ 40% dismissal rate)
- AI-assisted rule clarification drafting with impact what-if calculations showing affected clearance counts
- Audit log tracking every mutation, role action, dismissal, and clearance decision

3. The Three Round 3 Delivered Improvements

Improvement 1: Context-Aware Dependency Linking & Measured Precision Report

Committed: Prevent ambiguous or qualified statements from prematurely creating hard dependencies, isolate historical context from revalidations, assemble a gold benchmark evaluation set, and report real measured precision and recall on a public `/evaluation` page.

Delivered:

- Implemented deterministic context classification (`current`, `historical`, `quoted`, `hypothetical`) in `checker.ts`, ensuring past case studies and quotations never create dependencies on current facts.
- Built a `probable` links review queue in Review with dedicated `/api/findings/[id]/confirm-link` and `reject-link` endpoints, requiring explicit reviewer confirmation before ambiguous claims create clearance dependencies.
- Created a **50-case gold benchmark dataset** (40 detection cases and 10 revalidation scenarios in `tests/gold/`).
- Built the live `/evaluation` dashboard reporting measured real-time benchmark performance with confusion matrix breakdown and case filter inspector.

Benchmark Metric	Measured Value	Sample Size / Breakdown
Accuracy	`97.5%`	39 / 40 cases passed
Precision	`94.1%`	16 / 17 (TP=16, FP=1 on TC-028 term dep)
Recall	`100.0%`	16 / 16 (Zero false negatives)
F1 Score	`0.97`	N = 40 detection gold cases

Improvement 2: Visual Brand and Accessibility Checks for Images

Committed: Support visual asset sources in the source library (brand palettes, contrast thresholds), use Gemini vision to localize visual defect regions, and deterministically measure contrast ratios and palette compliance.

Delivered:

- Added first-class `asset` sources to the library (`brand_palette` with approved hex codes and Euclidean distance tolerance; `contrast_threshold` with 4.5:1 WCAG AA minimum).
- Built `src/server/services/imageChecker.ts` combining Gemini vision region proposals with deterministic mathematical verification using WCAG 2.1 relative luminance and Euclidean RGB distance.
- Created `ImageOverlay.tsx` with interactive SVG bounding boxes rendered directly over uploaded PNG/JPEG images (red blockers for contrast < 4.5:1, amber warnings for off-brand colors) with click-to-inspect findings.
- Added dedicated endpoints (`POST/GET /api/content/[id]/analyze-image`) and migration `20261113_image_checks.sql`.

Improvement 3: Governance Insights and Rule Clarification

Committed: Provide a `/insights` dashboard tracking at least 6 compliance metrics, an automated detector for unclear or frequently dismissed rules, and an AI-drafted clarification feature with what-if impact calculations.

Delivered:

- Built the `/insights` governance dashboard displaying 6 real-time metrics: Versions Cleared, Versions Pending, Median Clearance Time, Blocker counts, Warning counts, and Total Findings, alongside most-violated rules and top revalidation triggers.
- Implemented the unclear-rule detector identifying rules with ≥ 3 findings and ≥ 40% creator dismissal rates.

- Created `POST /api/insights/clarify` which drafts a clearer rule version via Gemini and deterministically calculates the "what-if" count of existing findings referencing that rule version.
- Integrated `/insights` into the primary top navigation.

4. Technical Architecture & Implementation Details

> **Stack:** Next.js 16.4.5 (App Router, Turbopack/Webpack) · TypeScript strict mode · Tailwind CSS tokenized design system · Neon Serverless PostgreSQL · Google Gemini API · Vitest · Playwright.

System Component Architecture:

- **Client Layer:** Clean, accessible React 19 UI with optimistic state updates, Toast notifications, RoleSwitcher demo control, and responsive drawer navigation.
- **API Handlers (`src/app/api/`):** Thin route handlers wrapping application services, extracting workspace sessions from cookie `pg_ws`, validating request payloads with Zod, and returning standardized error envelopes.
- **Domain Services (`src/server/services/`):** Encapsulates business logic for content analysis, document extraction, clearance state machines, revalidation dependency traversal, and source lifecycle operations.
- **Database Layer (`src/server/db/`):** PostgreSQL schema enforcing foreign keys, check constraints, unique version indices, and workspace isolation on every query (`WHERE workspace_id = $1`).

5. Testing & Verification Summary

Automated Unit Testing (53 tests passing across 9 test suites in Vitest):

- `tests/checker.test.ts` (13 tests): Verified numerical tolerance, unit conversion, date range checking, and context labeling.
- `tests/eval.test.ts` (4 tests): Genuine test run across all 40 gold detection cases asserting 97.5% accuracy, 94.1% precision, and 100% recall.
- `tests/clearance.test.ts` (6 tests): Verified clearance transactions, reviewer idempotency, and dependency pointer integrity.
- `tests/revalidate.test.ts` (5 tests): Verified selective revalidation, historical context immunity, and audit log generation.
- `tests/imageChecker.test.ts` (7 tests): Verified WCAG AA relative luminance, contrast ratio formulas, and Euclidean palette distance.
- `tests/insights.test.ts` (6 tests): Verified governance metric math and unclear-rule detection (≥ 40% dismissal threshold).
- `tests/sources.test.ts` (7 tests), `tests/comments.test.ts` (3 tests), `tests/roleChecks.test.ts` (2 tests).

End-to-End User Journey Testing: A full Playwright suite (`tests/e2e.spec.ts`) tests the entire lifecycle: ingesting Sample A, resolving blocker findings, human reviewer approval, superseding a pricing source in the library, and asserting that only dependent content reopens while historical content remains cleared.

Holdout Set Status: A holdout set structure was prepared at `tests/gold/holdout.json`, but was not run prior to submission to preserve test set integrity.

6. Product Roadmap & Future Direction

- **Phase 1: Real-Time CMS & Figma Integrations (Enterprise Marketing Teams):** Figma plugin and Headless CMS webhooks (Contentful, Sanity) to run ProofGate checks directly inside design files and draft stages before asset export.
- **Phase 2: Automated Visual Regression & Multipage PDF Scanning (Compliance Officers):** Client-side WebAssembly OCR engine (`tesseract.wasm`) for multi-page documents and scanned assets without text layers; structural logo aspect-ratio distortion measurement and trademark clear-space bounding checks.
- **Phase 3: Multi-Jurisdiction Regulatory Fact Sets (Legal & Regulated Brands):** Jurisdiction tags on sources (e.g., `IN`, `US`, `EU`) and claim tagging so marketing campaigns can be validated against regional regulatory

disclosures (e.g., ASCI guidelines in India, FTC disclosures in the US).